Katherine A. Bennett | Ph.D. Candidate

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Education

Johns Hopkins University — **Baltimore, MD**
Ph.D. in Planetary Science — *2022–2027 (anticipated)*

Wesleyan University — **Middletown, CT**
M.A. in Astronomy — *2020–2022*

University of Florida — **Gainesville, FL**
B.S./M.S. in Exercise Physiology — *2012–2018*

Publications (5 first author, 26 total publications)

First Author

Bennett, K. A., Gascón, C., Lustig-Yaeger, J., et al. (2026). "No Helium Detected in LHS 1140 b from Four JWST NIRISS/SOSS Transits", *in review at ApJL*

Bennett, K. A., Wiser, L. S., Tak, M. T., et al. (2026). "A JWST NIRISS/SOSS Transmission Spectrum of the Exo-Neptune HAT-P-11 b Reveals Aerosols and a High Metallicity In Agreement with the Mass-Metallicity Trend", *in review at AJ*

Bennett, K. A., MacDonald, R., Peacock, S., et al. (2025). "Additional JWST/NIRSpec Transits of the Rocky M Dwarf Exoplanet GJ 1132 b Reveal a Featureless Spectrum", *AJ, 170, 205*

Bennett, K. A., Sing, D. K., Stevenson, K. B., et al. (2025). "An HST Transmission Spectrum of the Closest M-Dwarf Transiting Rocky Planet LTT 1445Ab", *AJ, 169, 111*.

Bennett, K. A., Redfield, S., Oklopčić, A., et al. (2023). "Nondetection of Helium in the Hot Jupiter WASP-48b", *AJ, 165, 264*.

Second/Third Author

Bello-Arufe, A., Damiano, M., **Bennett, K. A.**, et al. (2025). "Evidence for a Volcanic Atmosphere on the sub-Earth L 98-59b", *ApJL, 980, 26*.

Lothringer, J. D., **Bennett, K. A.**, Sing, D. K., et al. (2025). "Refractory and Volatile Species in the UV-to-IR Transmission Spectrum of Ultra-hot Jupiter WASP-178b with HST and JWST", *AJ, 169, 274*.

May, E. M., MacDonald, R., **Bennett, K. A.**, et al. (2023). "Double Trouble: Two Divergent Transits of the Super-Earth GJ 1132 b Observed with JWST NIRSpec G395H", *ApJL, 959, L9*.

Contributing Author

Vermeiren, S. J., MacDonald, R. J., Huseby, L. et al. including **Bennett, K. A.**, et al. (2026). "Making Waves or Seeing Spots? Placing GJ 486 b on the Cosmic Shoreline with JWST NIRISS Transmission Spectroscopy", *in review at AJ*.

Dos Santos, L. A., Lothringer, J. D., Alam, M. K., **Bennett, K. A.**, et al. (2026). "Atmospheric Escape and Replenishment in Planetary Systems: The State of the Art in Theory and Observations", *in review at PASP*.

Wang, L. C., Mukherjee, S., Schmidt, S. P. et al. including **Bennett, K. A.** (2026). "Asymmetric Aerosol Distribution on the Terminators of the Warm Saturn WASP-69 b Revealed by JWST NIRISS/SOSS", *in review at ApJL*.

Stevenson, K. B., Fu, G., Sotzen, K. S. et al. including **Bennett, K. A.** (2026). "Evidence for LP 890-9d via Transit

Timing Variations", *in review at ApJL*.

Savel, A. B., Kempton, E. M. R., May, E. M. et al. including **Bennett, K. A.** (2026). "Precise Determination of the Metallicity and C/O of WASP-39 b From a Single JWST Instrument Mode with Phase-Resolved Cross-Correlation Retrievals", *in review at ApJL*.

Schmidt, S. P., May, E. M., Lothringer, J. D., et al. including **Bennett, K. A.** (2026). "Mitigating Charge Migration in JWST NIRISS Reveals That KELT-7 b is a Metal-enriched Ultra-hot Jupiter Orbiting a Young Metal-rich Star", *in review at AJ*.

Rustamkulov, Z., Peacock, S., Moran, S. E., et al. including **Bennett, K. A.** (2026). "Atmospheric Characterization of TRAPPIST-1 h and its Active Host Star with JWST/NIRSpec-PRISM", *accepted at AJ*.

Ashtari, R., Schmidt, S. P., Fu, G., et al. including **Bennett, K. A.** (2026). "A Clearer View of HAT-P-1 b: JWST NIRSpec G395H Reveals a Cloud-free Atmosphere of Water, Carbon Dioxide, and Possibly Hydrogen Sulfide", *AJ, 172, 275*.

Sing, D. K., Lothringer, J. D., Valenti, J. A. et al. including **Bennett, K. A.** (2026). "A JWST transiting survey of FGK stellar limb darkening: empirical evidence for quadratic laws and atmospheric model comparisons", *AJ, 172, 248*.

Mukherjee, S., Sing, D. K., Fu, G., et al. including **Bennett, K. A.** (2026). "Cloudy mornings and clear evenings on a giant extrasolar world", *Science, 392, 858*.

Batalha, N. E., Wallack, N., Gordon, T., et al. including **Bennett, K. A.** (2026). "JWST COMPASS Program: The 3–5 μm transmission spectrum of LTT 1445 A b", *AJ, 172, 52*.

Chachan, Y., Lothringer, J., Inglis, J., et al. including **Bennett, K. A.** (2025). "Strong NUV Refractory Absorption and Dissociated Water in the Hubble Transmission Spectrum of the Ultra Hot Jupiter KELT-20 b", *AJ, 170, 234*.

Fu, G., Mukherjee, S., Stevenson, K. B., et al. including **Bennett, K. A.** (2025). "Overcast mornings and clear evenings in hot Jupiter exoplanet atmospheres", *ApJL, 989, L17*.

Schmidt, S. P., MacDonald, R. J., Tsai, S., et al. including **Bennett, K. A.** (2025). "A Comprehensive Reanalysis of K2-18 b's JWST NIRISS+NIRSpec Transmission Spectrum", *AJ, 170, 298*.

Kirk, J., Stevenson, K. B., Fu, G., et al. including **Bennett, K. A.** (2024). "JWST/NIRCam Transmission Spectroscopy of the Nearby Sub-Earth GJ 341b", *AJ, 167, 90*.

Moran, S., Stevenson, K. B., Sing, D. K., et al. including **Bennett, K. A.** (2023). "High Tide or Rip-Tide on the Cosmic Shoreline? A Water-Rich Atmosphere or Stellar Contamination for the Warm Super-Earth GJ 486b from JWST Observations", *ApJL, 948, L11*.

Lustig-Yaeger, J., Fu, G., May, E. M., et al. including **Bennett, K. A.** (2023). "A JWST Transmission Spectrum of a Nearby Earth-Sized Exoplanet", *Nature Astronomy, 179L*.

Vassilakos, G., Lei, H., Yang, Y., et al. including **Bennett, K. A.** (2018). "Deletion of muscle IGF-I transiently impairs growth and progressively disrupts glucose homeostasis in male mice", *FASEB, 33, 1*.

Observing Proposals and Grants

Awarded $357K in competitive funding as PI; $399K total as PI/Co-I/Collaborator

JWST

PI: Cycle 4 GO Program 7251 (2025), $270K. "Does Our Closest M-Dwarf Rocky Neighbor Have An Atmosphere? We Need to Find Out.", *46.1 hours awarded*.

Co-I: (PI Joshua Lothringer), Cycle 5 GO Program 10290 (2026). "Mistaken Identity? Resolving Ultra-hot Jupiter NUV Absorption to Measure Refractory-to-Volatile Ratios", *9.8 hours awarded*.

Co-I: (PI David Sing), Cycle 5 GO Program 9709 (2026). "Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the mass-metallicity relation", *182.5 hours awarded*.

Collaborator: (PI Joost Wardenier), Cycle 4 GO Program 6978 (2025), $22K. "The dayside of the coolest

ultra-hot Jupiter, KELT-20b, resolved with eclipse mapping", *16.8 hours awarded*.

Co-I: (PIs Jacob Lustig-Yaeger & Kevin Stevenson), Cycle 4 GO Program 7073 (2025). "Charting the Cosmic Shoreline", *244.8 hours awarded*.

Co-I: (PI David Sing), Cycle 3 GO Program 5924 & HST Cycle 31 GO Program 17612 (2023). "JWST's Exoplanet Grand Tour Spectroscopic Survey", *JWST, 127.3 hours; HST, 24 orbits awarded*.

Co-I: (PI Ryan MacDonald), Cycle 3 GO Program 5866 (2023). "Oceans of Uncertainty: A Veritable Waterworld or a Desolate Wasteland?", *15.7 hours awarded*.

Co-I: (PI Laura Mayorga), Cycle 3 AR Program 4896 (2023), $20K. "AI-Enabled Modeling of Instrument Systematics for Exoplanet Time-Series Observations".

HST

PI: Cycle 33 GO Program 18140 (2025), $84K. "Flares, Everywhere and All at Once: An NUV/Optical Flare Study of Proxima Centauri", *23 orbits awarded*.

Co-I: (PI Joshua Lothringer), Cycle 34 GO Program 18140 (2026). "Mapping the Transition from Cloudy to Cloud-Free Atmospheres with NUV-Optical Transit Spectra of Three Giant Planets", *36 orbits awarded*.

Ground-Based Observations

co-PI: (with Zafar Rustamkulov), Apache Point Observatory 4Q2024 Program JH01:
"Chromosphere Monitoring of an Enigmatic Triple Red Dwarf System" *APO 3.5 m, 14 hours awarded*
"Complementary Flare Monitoring of the Enigmatic Triple Red Dwarf LTT 1445ABC" *APO 0.5 m, 14 nights awarded*

Co-I: (PI Kevin Schlaufman), Keck 1/KPF (2025A Program N072) + WIYN/NEID. "A Measurement of Primordial Stellar Obliquity in a Wide-separation Giant Planet System", *Keck + WIYN, 0.75 nights awarded*.

Grants

Best JHU EPS Department Seminar Presentation (2024-25), $2K

Student Travel Grant Recipient (2021), NASA CT Space Grant Consortium, $1K

Teaching and Mentorship

Teaching Academy Certificate, Johns Hopkins University. Spring 2025

- Participated in 25 hours of pedagogical seminars and trainings, culminating in a 12.5-hour teaching practicum:

Instructor, Department of Earth & Planetary Sciences, Johns Hopkins University. Fall 2024

- Independently developed curriculum for a 1-credit course titled *Earth is a Planet in Space* (*There Are 5,000 Others*).
- Led eight students through lecture + activity-based course. Assignments consisted of plotting exercises to develop real-world skills in data visualization.

Graduate Teaching Assistant, Astronomy Department, Wesleyan University. 2020–2022

- Led weekly office hours and evening telescope observing sessions.
- Met individually with students as needed. Graded ∼40 assignments weekly.
  - Introduction to Astrophysics (Fall 2020)
  - Conceptual Astronomy: Science Fact vs. Science Fiction (Spring 2021)
  - The Dark Side of the Universe (Fall 2021)
  - Observational Astronomy (Spring 2022)

Students Advised:

- Lily Kim, JHU EPS undergraduate student. Mars surface roughness research project.

Leadership and Service

- Board Member, Community Involvement Initiative, Rocky Worlds DDT Program — Sep. 2026 – present
  - Appointed by STScI to be responsible for orchestrating community involvement and coordination to optimize the scientific output of the Rocky Worlds Program
- Bromery Committee Member, Department of Earth & Planetary Sciences, JHU — Nov. 2023 – present
  - Identified speakers and organized logistics for weekly department Bromery seminar
- Reviewer: AAS Journals, MNRAS — December 2023 – present
- Science Organizing Committee Member, STScI/JHU ExoJamboree — Oct. – Nov. 2026
- Local Organizing Committee Member, Rocky Worlds DDT Workshop — Oct. – Nov. 2026
- Executive Secretary, NASA Astrophysics Review Panel — April 2026
- Judge, AAS Chambliss Student Poster Competition — January 2025
- Program Organizing Committee Member, Planet Characterization Conference 2024 — July 2023 – Feb 2024
- American Astronomical Society's Congressional Visits Day Participant (virtual) — March 2021

Honors and Awards

Best EPS Department Seminar Presentation 2024-25 ($2000 in research funds awarded) — May 2025

Nominee, JHU Career Impact Award — October 2024

Member of Phi Beta Kappa Society — March 2014

Florida Bright Futures Scholar — August 2012

Recipient of International Baccalaureate Diploma — June 2012

National Merit Scholarship Finalist — February 2012

Invited Talks

Consortium on Habitability and Atmospheres of M-dwarf Planets meeting (virtual) — October 2026

Is There Helium in the Atmosphere of LHS 1140 b?

Planetary Astronomy Lunch Seminar, University of Maryland, College Park, MD — December 2025

The Life of a Space Telescope (JWST's Version): Or, How We're Learning More About Rocky Worlds and Gas Giants Than Ever Before

Earth & Planets Laboratory Astro Seminar, Carnegie Science, Washington, D.C. — September 2024

The Hunt for Rocky Exoplanet Atmospheres: What We Have Learned (So Far) About Two Neighbors and Where to Go From Here

Planet Characterization Workshop, Lunar and Planetary Institute, Houston, TX — February 2024

Taxonomy from an Observer's Perspective: What We (Don't) Know About Planets Outside the Solar System

Planetary Astronomy Lunch Seminar, University of Maryland, College Park, MD — December 2023

Searching for M-Dwarf Rocky Planet Atmospheres with HST and JWST

Exoplanets and Disks Group Meeting, University of Amsterdam (virtual) — February 2023

Non-detection of Helium in the Extended Atmosphere of the Hot Jupiter WASP-48b

Contributed Talks and Posters

Earth & Planetary Sciences Department Seminar, JHU, Baltimore, MD — April 2026

What's that planet made of?! Constraints on the metal content of the warm Neptune HAT-P-11 b

STScI Atmospheric Escape and Replenishment Workshop, Baltimore, MD November 2025
The Power of Coupling Transmission and Emission in Observing Atmospheres: The Case of GJ 1132 b

ExoPAG32 Early Career Talk (virtual, pre-recorded due to maternity leave) August 2025
Four JWST Transits of GJ 1132 b Reveal This Spectrum Is the Flattest We Have Seen Yet (And Why This Matters)

Earth & Planetary Sciences Department Seminar, JHU, Baltimore, MD January 2025
Is There an Atmosphere on the Rocky Exoplanet GJ 1132b? (Spoiler: Probably Not)

AAS245 Research Presentation, National Harbor, MD January 2025
A Precise Transmission Spectrum of the Rocky M-dwarf Exoplanet GJ 1132b with Four JWST/NIRSpec Transits

STScI/JHU ExoJamboree, Baltimore, MD November 2024
Transmission Spectroscopy of Our Closest Transiting M-Dwarf Rocky Exoplanet Neighbor

Earth & Planetary Sciences Department Seminar, JHU, Baltimore, MD January 2024
Which Rocky Exoplanets Have Atmospheres? A Close Neighbor Offers a Glimpse of Hope

AAS243 Research Presentation, New Orleans, LA January 2024
An HST Reconnaissance Study of the Closest M-Dwarf Transiting Rocky Planet

Sagan Exoplanet Summer Workshop, Pasadena, CA (poster) July 2023
A Tale of Two Data Reductions: The JWST Transmission Spectrum of the Warm Super-Earth GJ 1132b

Exoclimes VI, Exeter, UK (poster) June 2023
A Tale of Two Data Reductions: The JWST Transmission Spectrum of the Warm Super-Earth GJ 1132b

STScI Spring Symposium, Baltimore, MD May 2023
Two Possible Interpretations of the JWST Transmission Spectrum of the Warm Super-Earth GJ 486b

Earth & Planetary Sciences Department Seminar, JHU, Baltimore, MD May 2023
Searching for Atmospheres on Rocky Worlds with JWST

AAS241 Research Presentation, Seattle, WA January 2023
Non-detection of Helium in the Extended Atmosphere of the Hot Jupiter WASP-48b

Astronomy Department Seminar, Wesleyan University, Middletown, CT May 2022
Searching for Helium in the Extended Atmosphere of the Hot Jupiter WASP-48b

CHAMPs Exoplanet Early Career Highlight Seminar (virtual) January 2022
Searching for Helium in the Extended Atmosphere of the Hot Jupiter WASP-48b

Graduate Speaker Series, Wesleyan University, Middletown, CT December 2021
Exploring Other Worlds: Searching for Helium in the Atmosphere of a Hot Jupiter

Outreach

Moderator, Conversation with Hakeem Oluseyi, JHU Icons of Impact Series, Baltimore, MD November 2024
Exoplanets Talk, QuarkNet Collaboration, Baltimore, MD July 2024

Exoplanets Talk, Hereford High School Science National Honor Society, Hereford, MD — May 2024
Science Outreach Talk, Baltimore Polytechnical Institute, Baltimore, MD — November 2023
Speaker, Earth & Planetary Sciences Geosciences Camp, Baltimore, MD — August 2023
Guest Lecturer, STScI Public Lecture Series (virtual) — April 2023
Astronomy Outreach Talk, Baltimore Montessori Public Charter School, Baltimore, MD — March 2023
Science Olympiad Coach, Fallstaff Elementary School, Baltimore, MD — Spring 2023
Astronomy Outreach Event, Hanmer Elementary School, Wethersfield, CT — May 2022
Panelist, STEAMfest, Housatonic Community College, Bridgeport, CT — November 2021
Sidewalk Science with Russell Library, Middletown, CT — September 2021

Media Appearances and Quotations

"GJ 1132 b Doesn't Have An Atmosphere, According To New JWST Data", *Universe Today* — August 2025
"JWST's Next Year of Amazing Science Revealed", *Scientific American* — April 2025
Appearance on PBS NOVA Episode "New Eye on the Universe" — February 2023

Past Professional Experience

University of Florida Health Cardiac Rehabilitation, Gainesville, FL — June 2018 – April 2020

- Directed ∼20 patients daily through guided exercise sessions with a small three-person team
- Mentored patients on exercise and nutrition education
- Trained a total of eight interns and coordinated treatment plans with attending physicians